MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE EBD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic inverter module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0814 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBD.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/197-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

* 3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

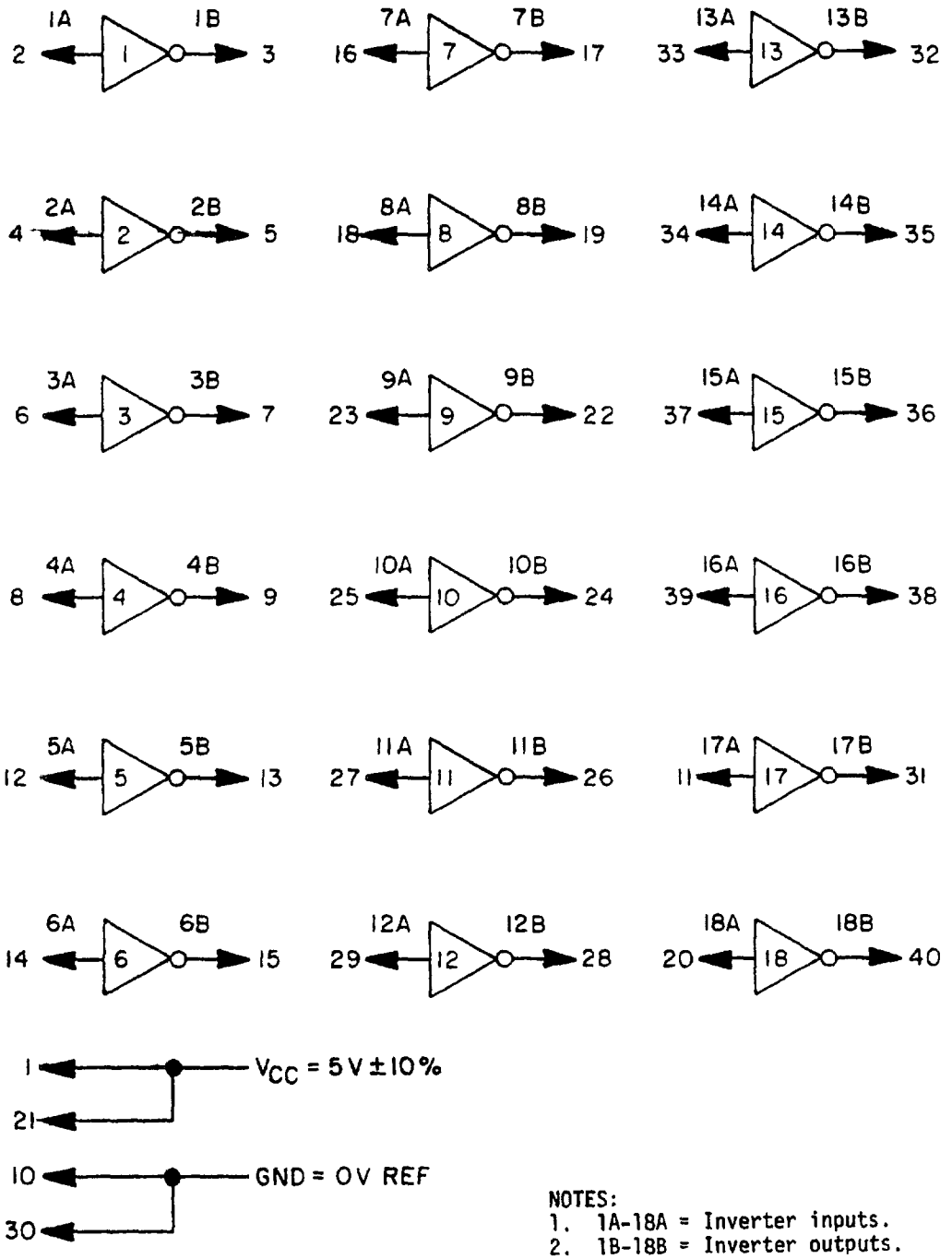
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-18A = Inverter inputs.
2. 1B-18B = Inverter outputs.

* FIGURE 1. EBD module function diagram, inverter.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B18	V_{OUT} (low loaded)	IV	S	0.01	1.40	0.01	1.40	0.01	1.40	0.01	1.40	V
C1-C18	Input breakdown currrent	V	S	0	1	0	1	0	1	0	1	mA
D1-D18	Input clamp diode voltage	VI	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E18	Output voltage, low level, I_{OL} = 16 mA	VII	X	0	400	0	400	0	400	0	400	mV
F1-F18	Output voltage, low level, I_{OL} = 30 mA	VIII	X	0	700	0	700	0	700	0	700	mV
G1-G18	Output leakage current	IX	X	0	250	0	250	0	250	0	250	µA
H1-H18	Input current, high level	X	X	0	40	0	40	0	40	0	40	µA
J1-J18	Input current, low level	XI	X	-0.20	-1.73	-0.20	-1.73	-0.20	-1.73	-0.20	-1.73	mA
K1,K2	Continuity	XII	X	0	1	0	1	0	1	0	1	Ω
K3,K4	Continuity	XII	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
L1	Power supply current, low level outputs	XIII	S	48	153	48	153	48	153	48	153	mA
L2	Power supply current, high level outputs	XIII	S	45	144	45	144	45	144	45	144	mA
M1-M18	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	4	45	4	45	4	50	4	50	ns
M19-M36	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	10	90	10	90	10	100	10	100	ns
N1-N18	Transition time (t_{THL}), C_L = 50 pF	XIV	S	1	10	1	10	1	12	1	12	ns
N19-N36	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	1	15	1	15	1	17	1	17	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1-P18	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	4	40	4	40	4	42	4	42	ns
P19-P36	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	8	65	8	65	8	67	8	67	ns
Q1-Q18	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2	35	2	35	2	37	2	37	ns
Q19-Q36	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4	50	4	50	4	52	4	52	ns
R1-R18	Output crosstalk, low level	XVI	Q	0	1,250	0	1,250	0	1,250	0	1,250	mV
R19-R36	Output crosstalk, low level	XVI	Q	0	500	0	500	0	500	0	500	mV
S1-S18	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XVIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
V1-V18	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/197-1 supersedes part number M28787/197. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	
All outputs at low level	842 mW (outputs not loaded)
All outputs at high level	792 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2 V minimum, V_{CC} maximum

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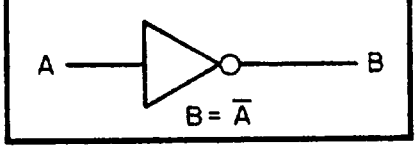
Input current, low level	
V_{IL} = 0 V	-1.73 mA maximum
V_{IL} = 0.4 V	-1.6 mA maximum
Input current, high level	
V_{IH} = 2.4 V	40 µA maximum
V_{IH} = 5.5 V	1 mA maximum
Output voltage, low level	
I_{OL} = 16 mA	0.4 V maximum
I_{OL} = 30 mA	0.7 V maximum
Output voltage, high level	
Applied voltage	30 V maximum
Output leakage current	
V_{OH} = 30 V	250 µA maximum
Output drive current maximum capability	
V_{OL} = 0.4 V	16 mA
V_{OL} = 0.7 V	30 mA

* 6.5 Module function. This module contains independent open collector high-voltage inverters.

6.5.1 Input to output relationship. Each inverter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ 2/

INPUT	OUTPUT
A	B
L	H
H	L



1/ H = High level.
2/ L = Low level.

* 6.6 Power pins. All electrical parameters are measured with pins 1 and 21 connected separately to V_{CC} and pins 10 and 30 connected separately to 0 volt return. System problems may result from a different power configuration.

* 6.7 Crosstalk relationships. The module has the following crosstalk relationships.

6.7.1 Module crosstalk. When connected in the worst case mode of operation (all but one gate switching simultaneously from high level to low level), the EBD module can exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as a 1,250 mV positive going spike coincident with the negative edge of the switching outputs.

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6.7.2 Circuit crosstalk. The crosstalk noise present on any output is a function of the number of simultaneously switching outputs. The noise generated on a quiet low-level output of a circuit in group A (circuits 1 through 6), group B (circuits 7 through 12), or group C (circuits 13 through 18) can be as high as 150 mV for each output switching in that group. The noise generated by gates outside the group under consideration can be as high as 42 mV for each switching output. Crosstalk will exceed these usage guidelines if the module is used for lamp switching, relay switching or high drive (30 mA) logic switching.

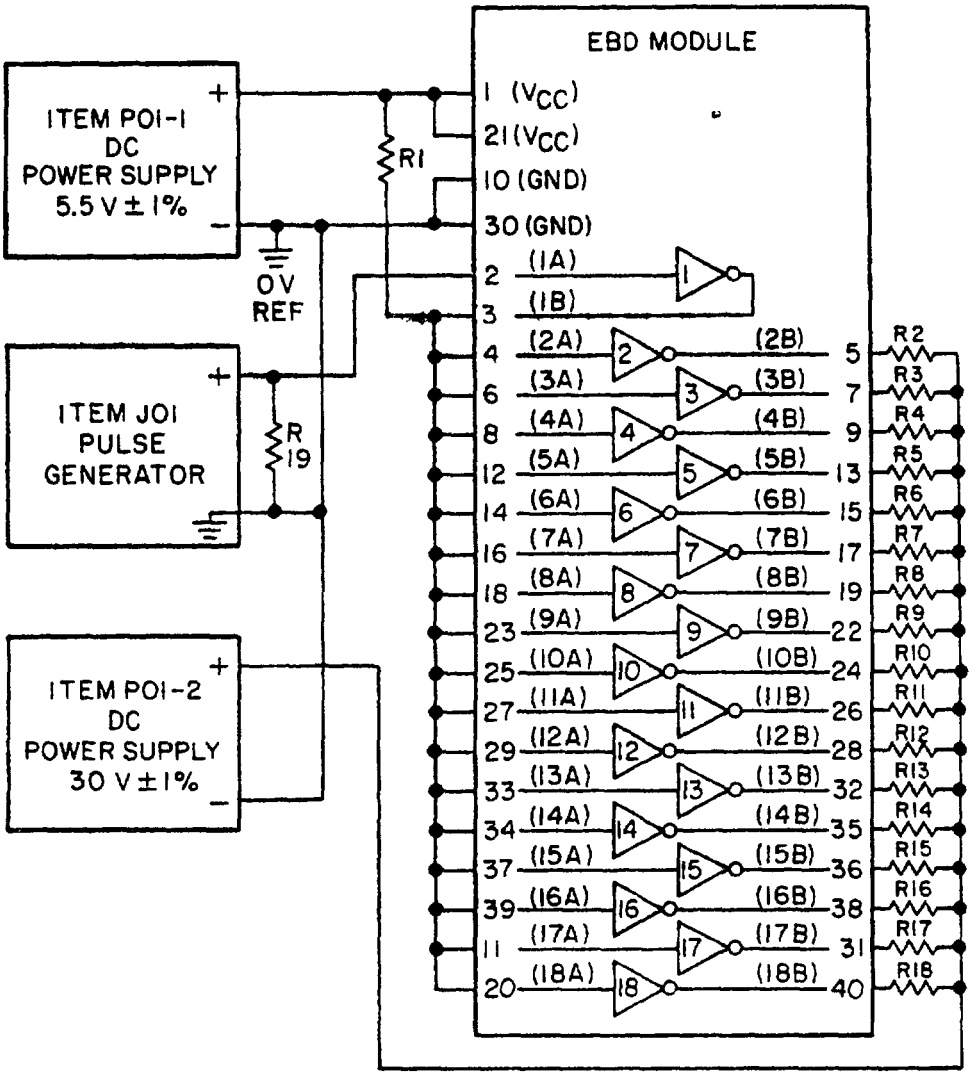
6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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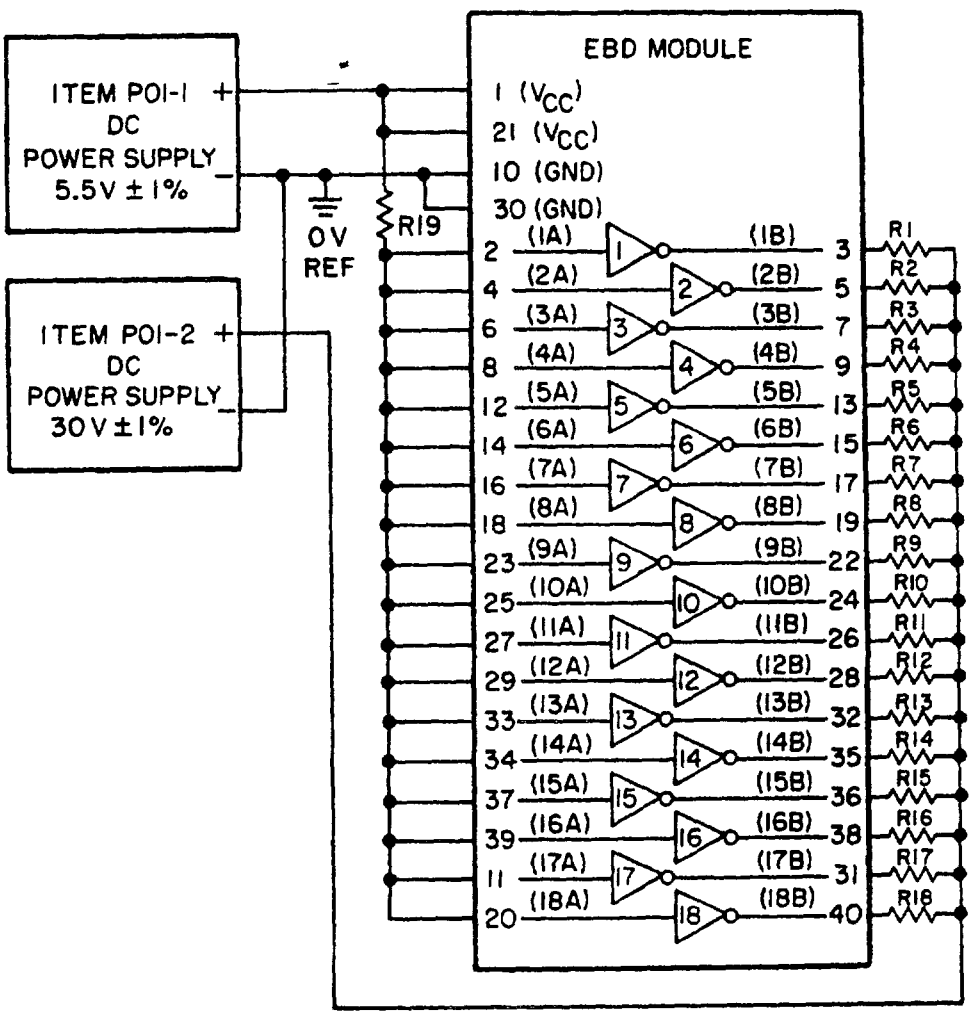
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NOTES:
1. R1 = 1.74 kΩ ±1%, 1/8 W.
2. R2-R18 = 976Ω ±1%, 1 W.
3. R19 = 51Ω ±5%, 1/4 W.
4. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R19 = 976Ω ±1%, 1 W.

FIGURE 3. Component temperature test circuit

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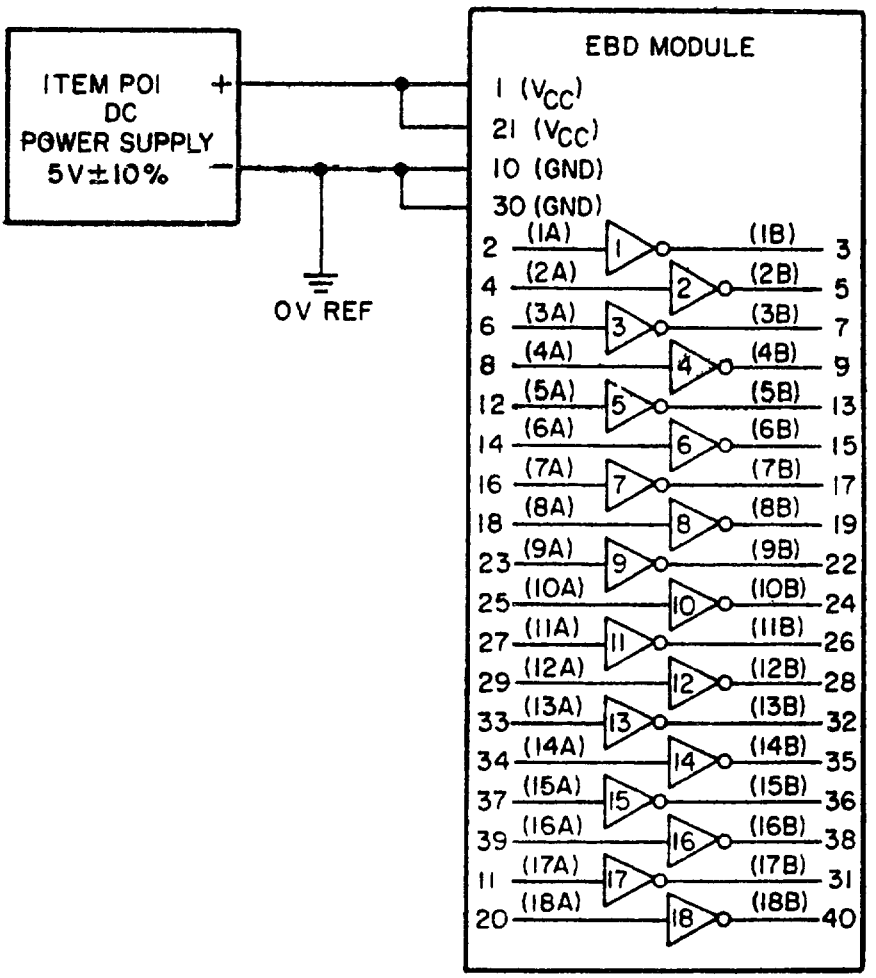


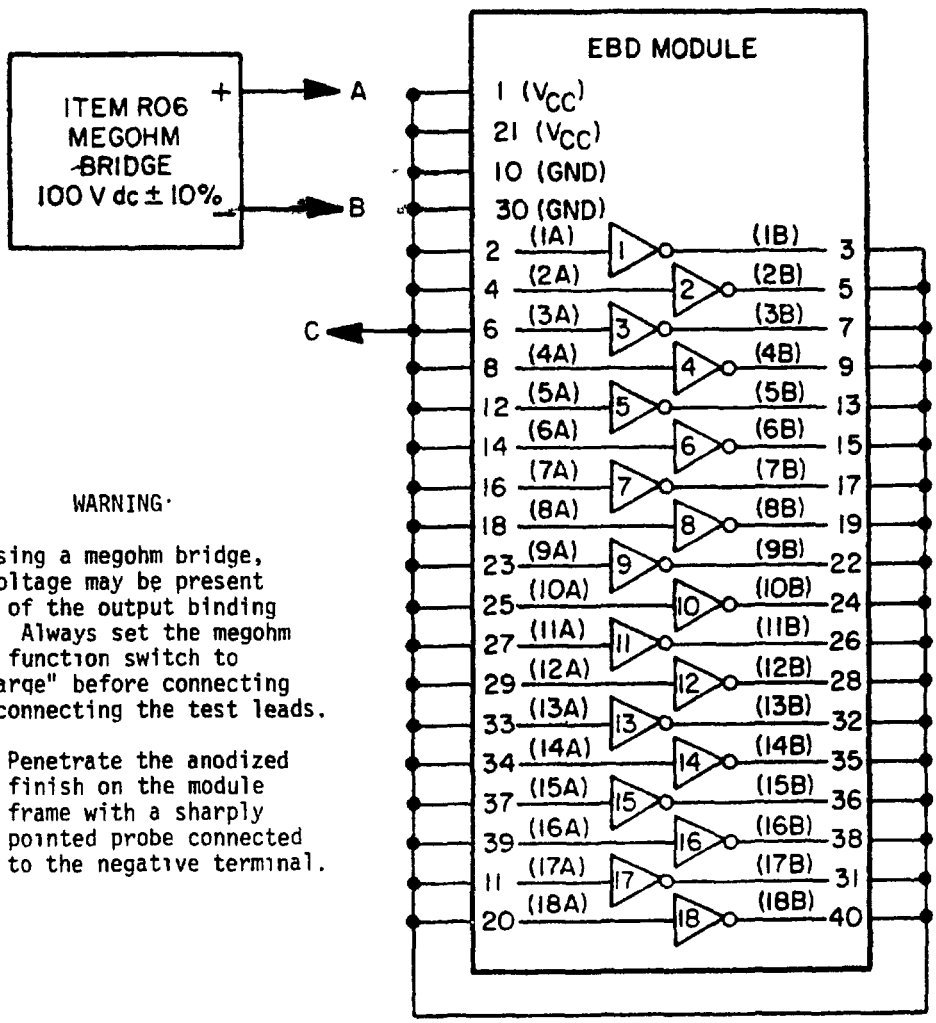
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING·

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

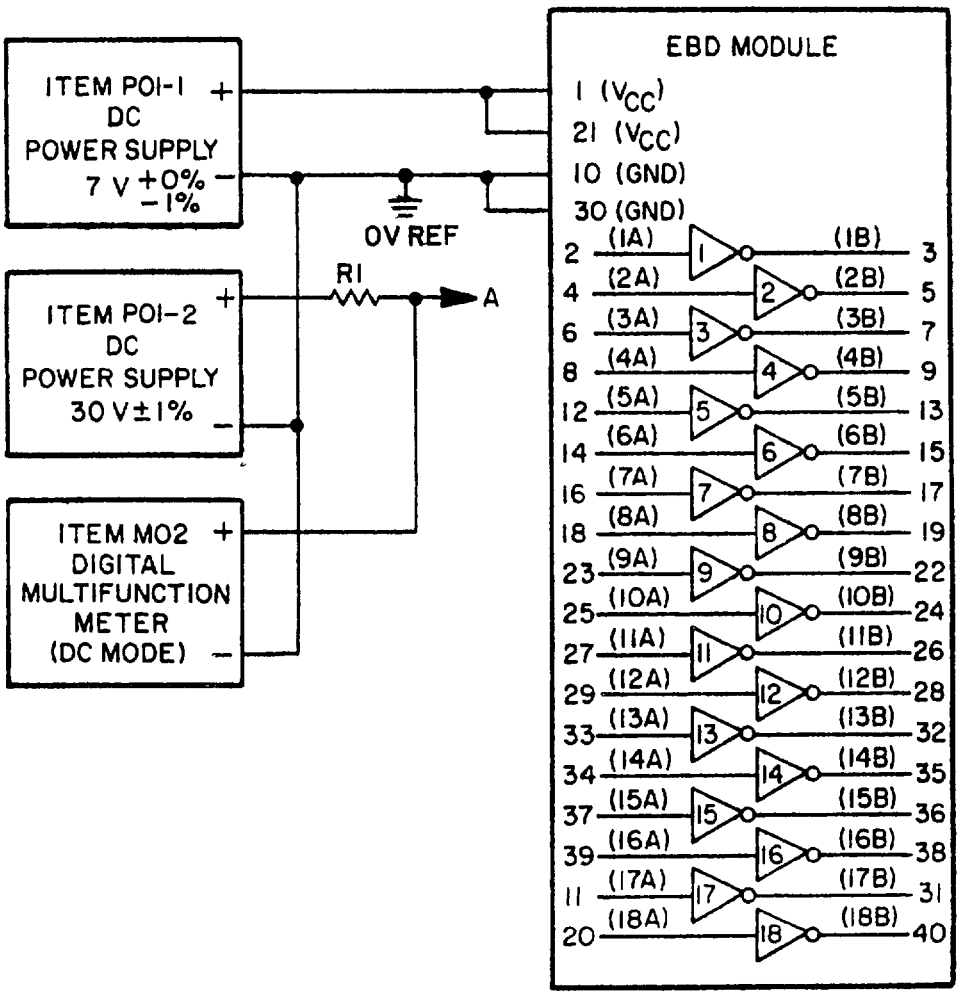
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1B V_{OUT} (low loaded)	3	0.01	1.40	V	6
B2	2B "	5	"	"	"	"
B3	3B "	7	"	"	"	"
B4	4B "	9	"	"	"	"
B5	5B "	13	"	"	"	"
B6	6B "	15	"	"	"	"
B7	7B "	17	"	"	"	"
B8	8B "	19	"	"	"	"
B9	9B "	22	"	"	"	"
B10	10B "	24	"	"	"	"
B11	11B "	26	"	"	"	"
B12	12B "	28	"	"	"	"
B13	13B "	32	"	"	"	"
B14	14B "	35	"	"	"	"
B15	15B "	36	"	"	"	"
B16	16B "	38	"	"	"	"
B17	17B "	31	"	"	"	"
B18	18B "	40	"	"	"	"

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NOTE: R1 = 976Ω ±1%, 1 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

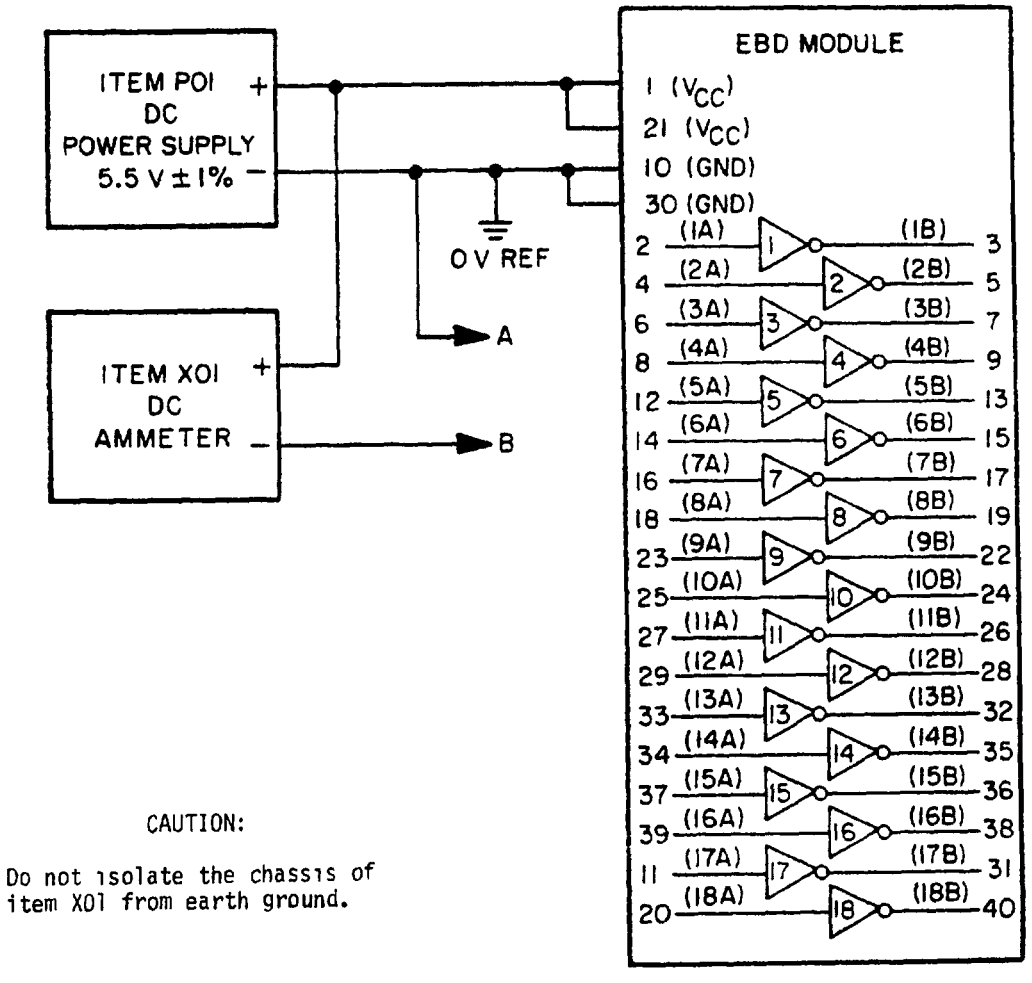
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	4,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	2	0	1	mA	7
C2	2A "	2,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	4	"	"	"	"
C3	3A "	2,4,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	6	"	"	"	"
C4	4A "	2,4,6,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	8	"	"	"	"
C5	5A "	2,4,6,8,11,14,16,18,20, 23,25,27,29,33,34,37,39	12	"	"	"	"
C6	6A "	2,4,6,8,11,12,16,18,20, 23,25,27,29,33,34,37,39	14	"	"	"	"
C7	7A "	2,4,6,8,11,12,14,18,20, 23,25,27,29,33,34,37,39	16	"	"	"	"
C8	8A "	2,4,6,8,11,12,14,16,20, 23,25,27,29,33,34,37,39	18	"	"	"	"
C9	9A "	2,4,6,8,11,12,14,16,18, 20,25,27,29,33,34,37,39	23	"	"	"	"
C10	10A "	2,4,6,8,11,12,14,16,18, 20,23,27,29,33,34,37,39	25	"	"	"	"
C11	11A "	2,4,6,8,11,12,14,16,18, 20,23,25,29,33,34,37,39	27	"	"	"	"
C12	12A "	2,4,6,8,11,12,14,16,18, 20,23,25,27,33,34,37,39	29	"	"	"	"
C13	13A "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,34,37,39	33	"	"	"	"
C14	14A "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,37,39	34	"	"	"	"
C15	15A "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,39	37	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C16	16A Input breakdown current	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,37	39	0	1	mA	7
C17	17A "	2,4,6,8,12,14,16,18,20, 23,25,27,29,33,34,37,39	11	"	"	"	"
C18	18A "	2,4,6,8,11,12,14,16,18, 23,25,27,29,33,34,37,39	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XOI from earth ground.

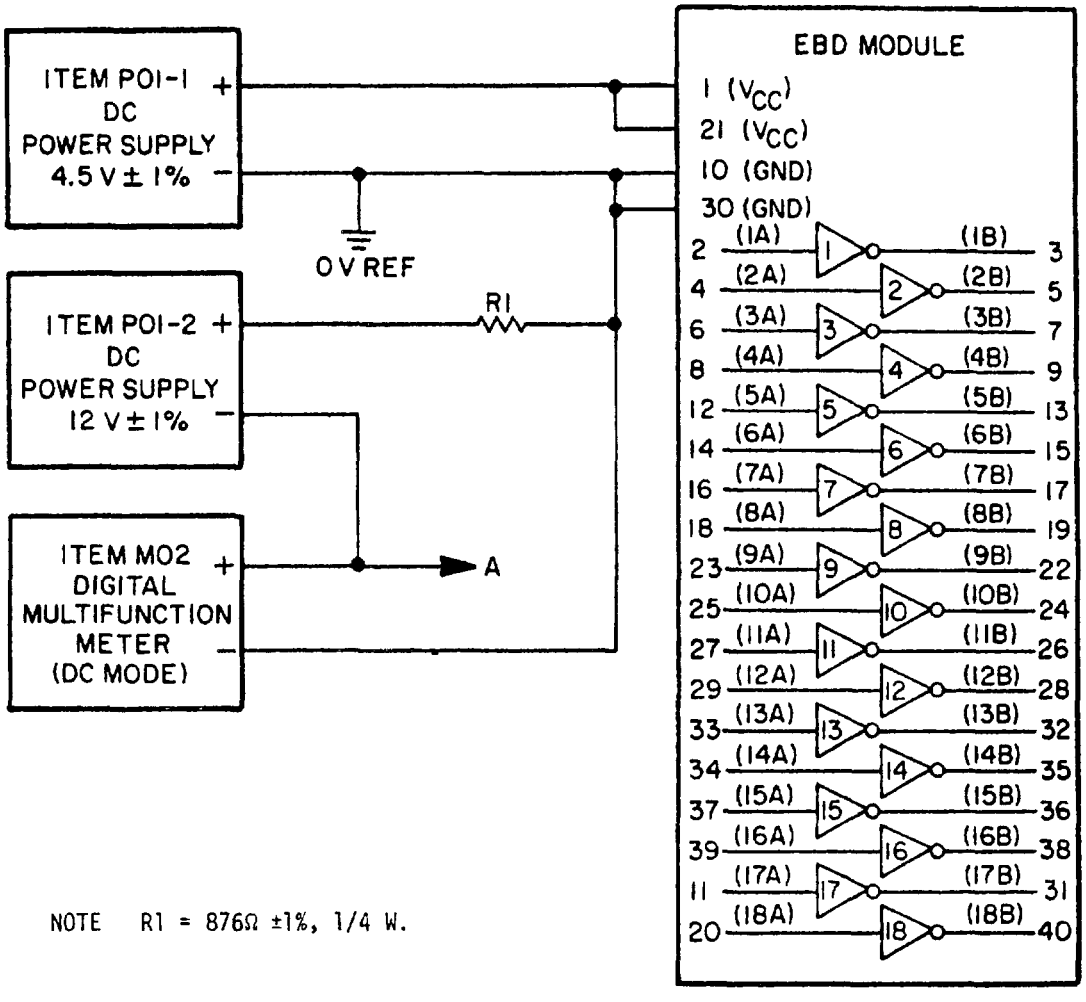
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1A Input clamp diode voltage	2	-0.4	-1.5	V	8
D2	2A "	4	"	"	"	"
D3	3A "	6	"	"	"	"
D4	4A "	8	"	"	"	"
D5	5A "	12	"	"	"	"
D6	6A "	14	"	"	"	"
D7	7A "	16	"	"	"	"
D8	8A "	18	"	"	"	"
D9	9A "	23	"	"	"	"
D10	10A "	25	"	"	"	"
D11	11A "	27	"	"	"	"
D12	12A "	29	"	"	"	"
D13	13A "	33	"	"	"	"
D14	14A "	34	"	"	"	"
D15	15A "	37	"	"	"	"
D16	16A "	39	"	"	"	"
D17	17A "	11	"	"	"	"
D18	18A "	20	"	"	"	"

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NOTE R1 = 876Ω ±1%, 1/4 W.

* FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Output voltage (low-level) test.

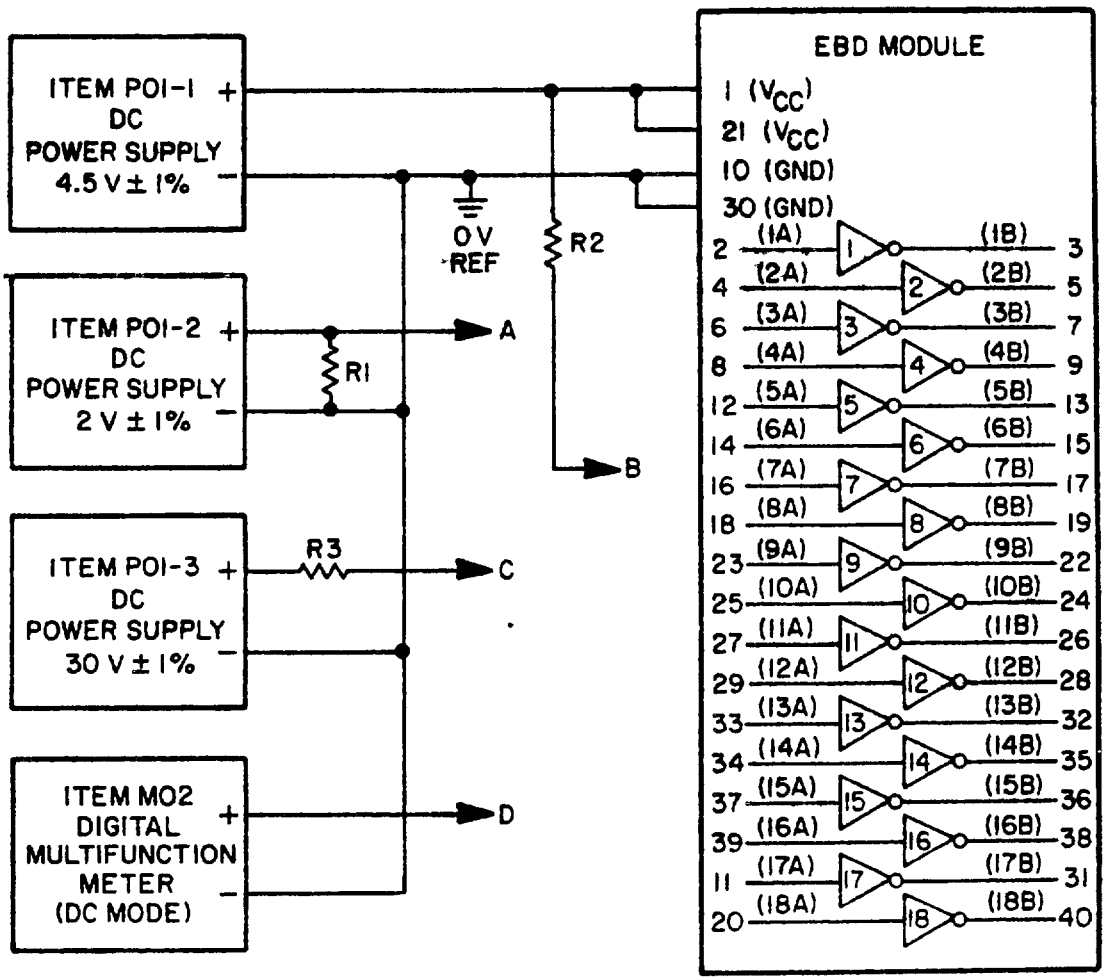
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1B Output voltage, low level, I_{OL} = 16 mA	2	3	Open	3	0	400	mV	9
E2	2B "	4	5	"	5	"	"	"	"
E3	3B "	6	7	"	7	"	"	"	"
E4	4B "	8	9	"	9	"	"	"	"
E5	5B "	12	13	"	13	"	"	"	"
E6	6B "	14	15	"	15	"	"	"	"
E7	7B "	16	17	"	17	"	"	"	"
E8	8B "	18	19	"	19	"	"	"	"
E9	9B "	23	22	"	22	"	"	"	"
E10	10B "	25	24	"	24	"	"	"	"
E11	11B "	27	26	"	26	"	"	"	"
E12	12B "	29	28	"	28	"	"	"	"
E13	13B "	33	32	"	32	"	"	"	"
E14	14B "	34	35	"	35	"	"	"	"
E15	15B "	37	36	"	36	"	"	"	"
E16	16B "	39	38	"	38	"	"	"	"
E17	17B "	11	31	"	31	"	"	"	"
E18	18B "	20	40	"	40	"	"	"	"

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TABLE VIII. Output voltage (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1B Output voltage, low level, I_{OL} = 30 mA	2	Open	3	3	0	700	mV	9
F2	2B "	4	"	5	5	"	"	"	"
F3	3B "	6	"	7	7	"	"	"	"
F4	4B "	8	"	9	9	"	"	"	"
F5	5B "	12	"	13	13	"	"	"	"
F6	6B "	14	"	15	15	"	"	"	"
F7	7B "	16	"	17	17	"	"	"	"
F8	8B "	18	"	19	19	"	"	"	"
F9	9B "	23	"	22	22	"	"	"	"
F10	10B "	25	"	24	24	"	"	"	"
F11	11B "	27	"	26	26	"	"	"	"
F12	12B "	29	"	28	28	"	"	"	"
F13	13B "	33	"	32	32	"	"	"	"
F14	14B "	34	"	35	35	"	"	"	"
F15	15B "	37	"	36	36	"	"	"	"
F16	16B "	39	"	38	38	"	"	"	"
F17	17B "	11	"	31	31	"	"	"	"
F18	18B "	20	"	40	40	"	"	"	"

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NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. R2 = 255Ω ±1%, 1/8 W.
3. R3 = 976Ω ±1%, 1 W.

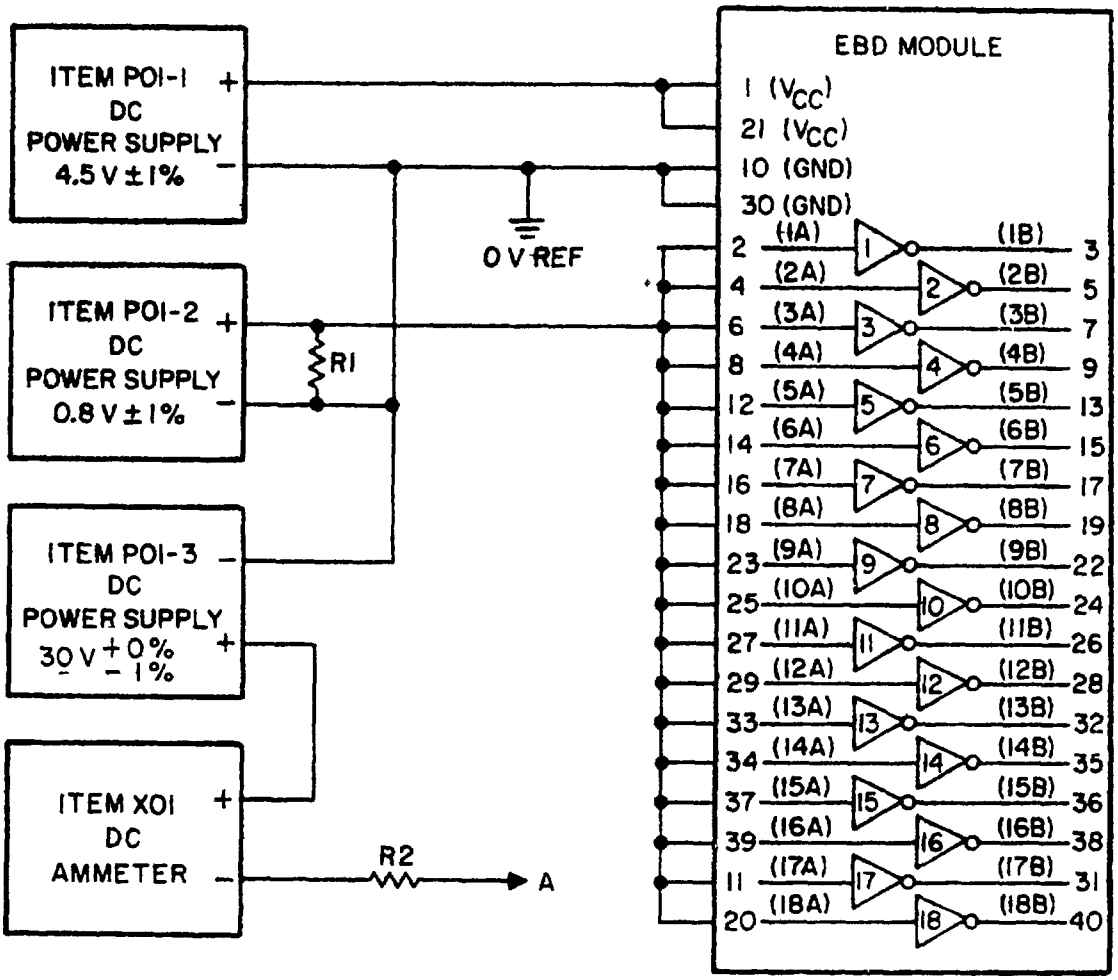
* FIGURE 9. Output voltage (low-level) test circuit for tables VII and VIII.

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TABLE IX. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
G1	1B Output leakage current	3	0	250	µA	10
G2	2B "	5	"	"	"	"
G3	3B "	7	"	"	"	"
G4	4B "	9	"	"	"	"
G5	5B "	13	"	"	"	"
G6	6B "	15	"	"	"	"
G7	7B "	17	"	"	"	"
G8	8B "	19	"	"	"	"
G9	9B "	22	"	"	"	"
G10	10B "	24	"	"	"	"
G11	11B "	26	"	"	"	"
G12	12B "	28	"	"	"	"
G13	13B "	32	"	"	"	"
G14	14B "	35	"	"	"	"
G15	15B "	36	"	"	"	"
G16	16B "	38	"	"	"	"
G17	17B "	31	"	"	"	"
G18	18B "	40	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XOI from earth ground.

NOTES:
1. R1 = 10Ω ±5%, 1/4 W.
2. R2 = 976Ω ±1%, 1 W.
3. Connect all pins as specified before connecting or disconnecting point A.

FIGURE 10. Output leakage current test circuit for table IX.

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TABLE X. Input current (high-level) test.

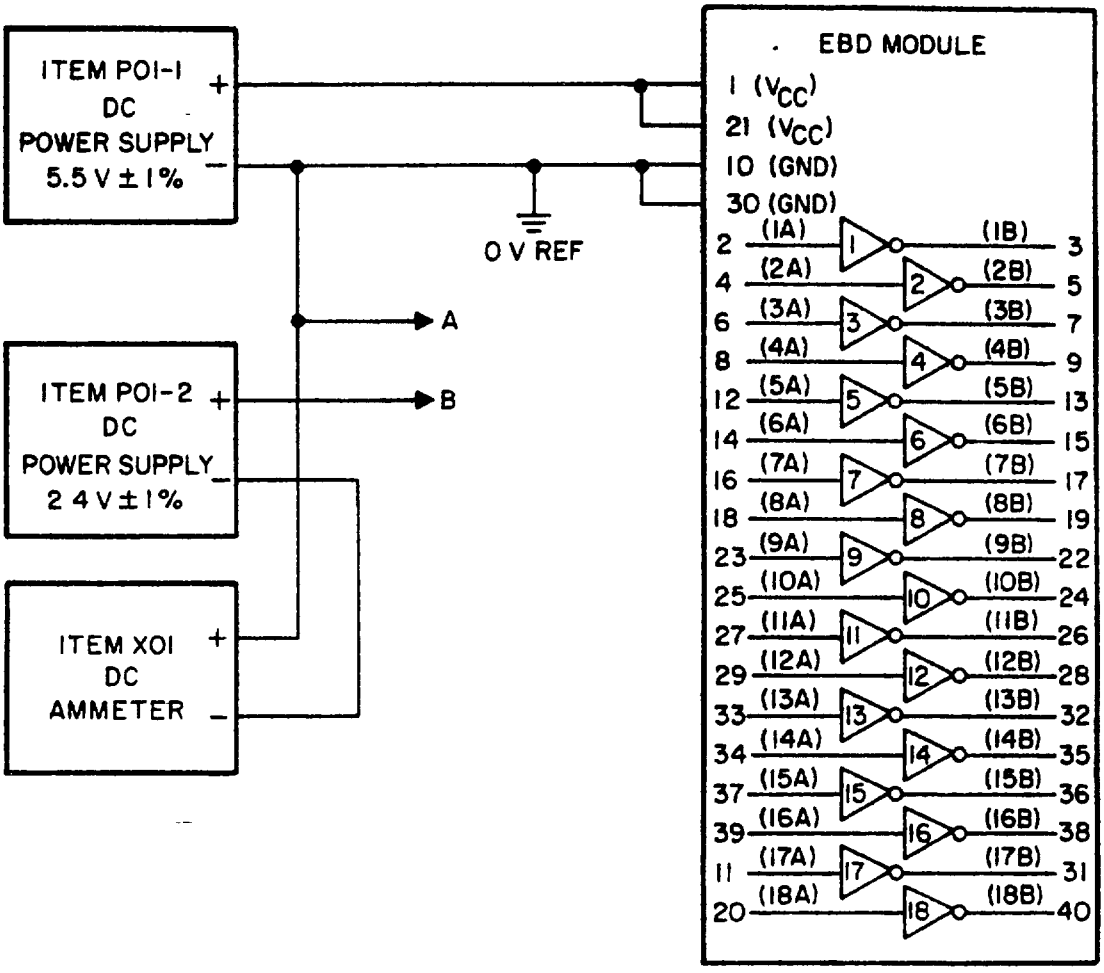
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	A1 Input current, high level	4,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	2	0	40	µA	11
H2	A2 "	2,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	4	"	"	"	"
H3	A3 "	2,4,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	6	"	"	"	"
H4	A4 "	2,4,6,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	8	"	"	"	"
H5	A5 "	2,4,6,8,11,14,16,18,20, 23,25,27,29,33,34,37,39	12	"	"	"	"
H6	A6 "	2,4,6,8,11,12,16,18,20, 23,25,27,29,33,34,37,39	14	"	"	"	"
H7	A7 "	2,4,6,8,11,12,14,18,20, 23,25,27,29,33,34,37,39	16	"	"	"	"
H8	A8 "	2,4,6,8,11,12,14,16,20, 23,25,27,29,33,34,37,39	18	"	"	"	"
H9	A9 "	2,4,6,8,11,12,14,16,18, 20,25,27,29,33,34,37,39	23	"	"	"	"
H10	A10 "	2,4,6,8,11,12,14,16,18, 20,23,27,29,33,34,37,39	25	"	"	"	"
H11	A11 "	2,4,6,8,11,12,14,16,18, 20,23,25,29,33,34,37,39	27	"	"	"	"
H12	A12 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,33,34,37,39	29	"	"	"	"
H13	A13 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,34,37,39	33	"	"	"	"
H14	A14 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,37,39	34	"	"	"	"
H15	A15 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,39	37	"	"	"	"

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H16	A16 Input current, high level	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,37	39	0	40	µA	11
H17	A17 "	2,4,6,8,12,14,16,18,20, 23,25,27,29,33,34,37,39	11	"	"	"	"
H18	A18 "	2,4,6,8,11,12,14,16,18, 23,25,27,29,33,34,37,39	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XOI from earth ground.

FIGURE 11. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

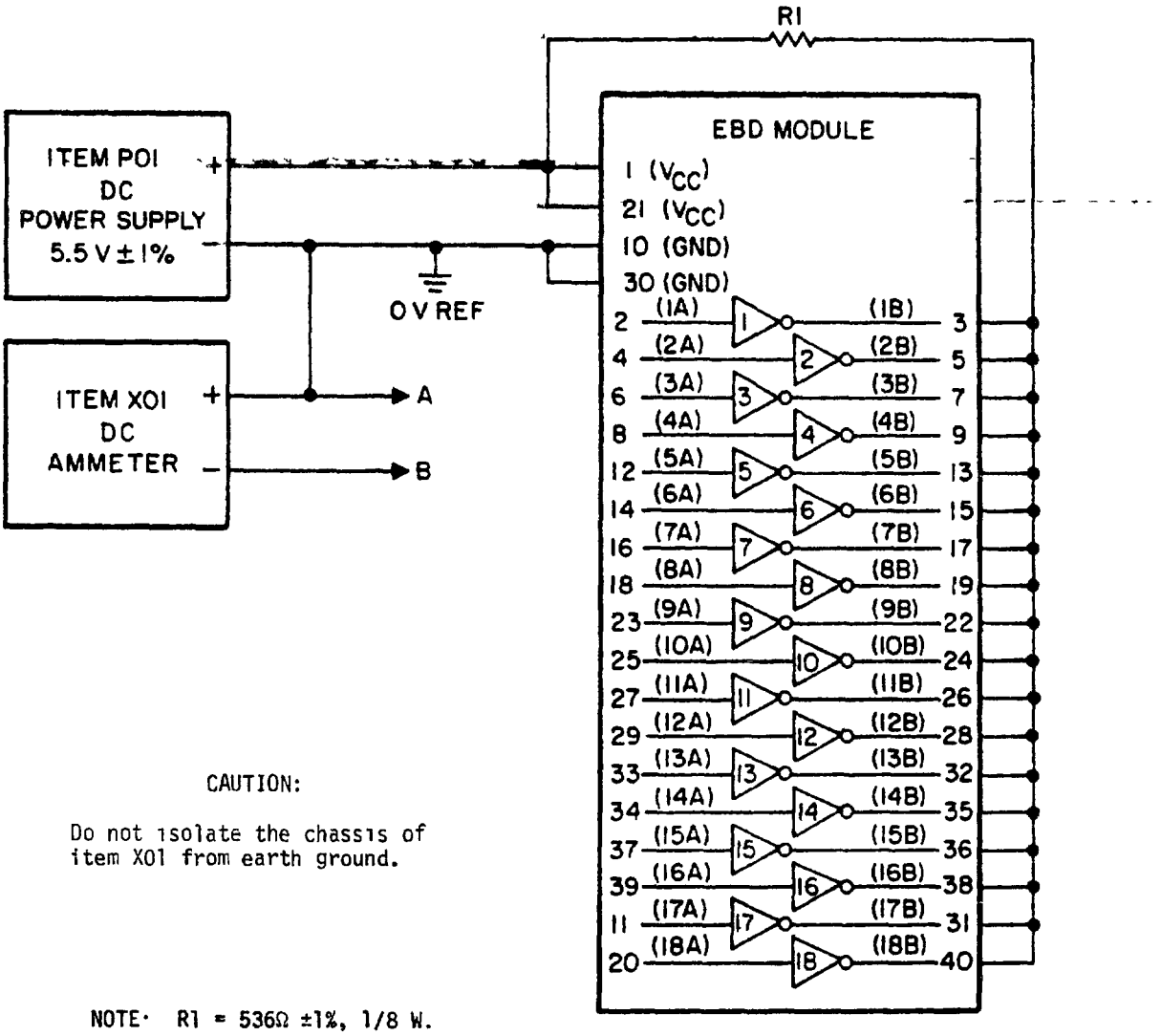
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	A1 Input current, low level	4,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	2	-0.20	-1.73	mA	12
J2	A2 "	2,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	4	"	"	"	"
J3	A3 "	2,4,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	6	"	"	"	"
J4	A4 "	2,4,6,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	8	"	"	"	"
J5	A5 "	2,4,6,8,11,14,16,18,20, 23,25,27,29,33,34,37,39	12	"	"	"	"
J6	A6 "	2,4,6,8,11,12,16,18,20, 23,25,27,29,33,34,37,39	14	"	"	"	"
J7	A7 "	2,4,6,8,11,12,14,18,20, 23,25,27,29,33,34,37,39	16	"	"	"	"
J8	A8 "	2,4,6,8,11,12,14,16,20, 23,25,27,29,33,34,37,39	18	"	"	"	"
J9	A9 "	2,4,6,8,11,12,14,16,18, 20,25,27,29,33,34,37,39	23	"	"	"	"
J10	A10 "	2,4,6,8,11,12,14,16,18, 20,23,27,29,33,34,37,39	25	"	"	"	"
J11	A11 "	2,4,6,8,11,12,14,16,18, 20,23,25,29,33,34,37,39	27	"	"	"	"
J12	A12 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,33,34,37,39	29	"	"	"	"
J13	A13 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,34,37,39	33	"	"	"	"
J14	A14 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,37,39	34	"	"	"	"
J15	A15 "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,39	37	"	"	"	"

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J16	A16 Input current, low level	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,37	39	-0.20	-1.73	mA	12
J17	A17 "	2,4,6,8,12,14,16,18,20, 23,25,27,29,33,34,37,39	11	"	"	"	"
J18	A18 "	2,4,6,8,11,12,14,16,18, 23,25,27,29,33,34,37,39	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: R1 = 536Ω ±1%, 1/8 W.

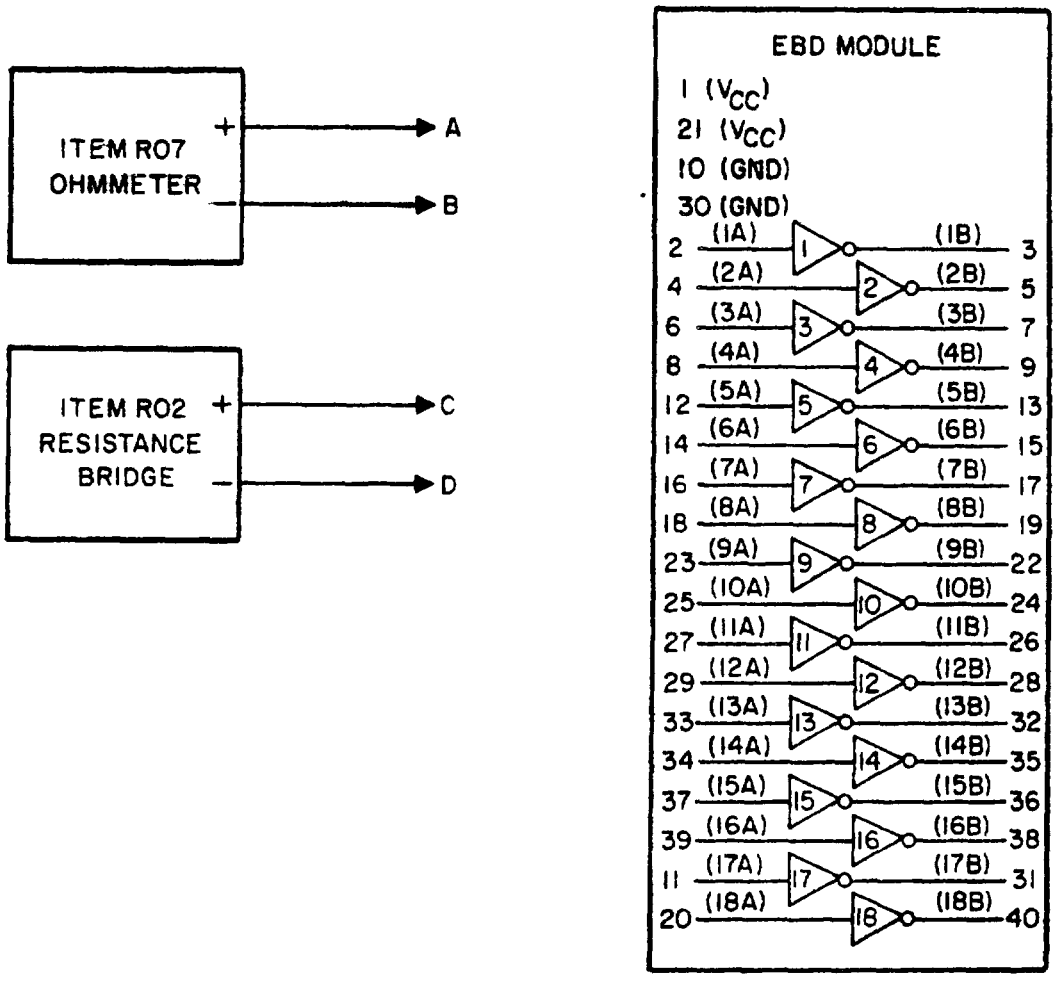
FIGURE 12. Input current (low-level) test circuit for table XI.

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TABLE XII. Continuity test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	Continuity	1	21	Open	Open	0	1	Ω	13
K2	"	10	30	Open	Open	0	1	"	"
K3	"	Open	Open	1	21	0.0	0.1	"	"
K4	"	Open	Open	10	30	0.0	0.1	"	"

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* FIGURE 13. Continuity test circuit for table XII.

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TABLE XIII. Power supply current (low-level and high-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Power supply current, low level outputs	Open	C	48	153	mA	14
L2	Power supply current, high level outputs	C	Open	45	144	mA	14

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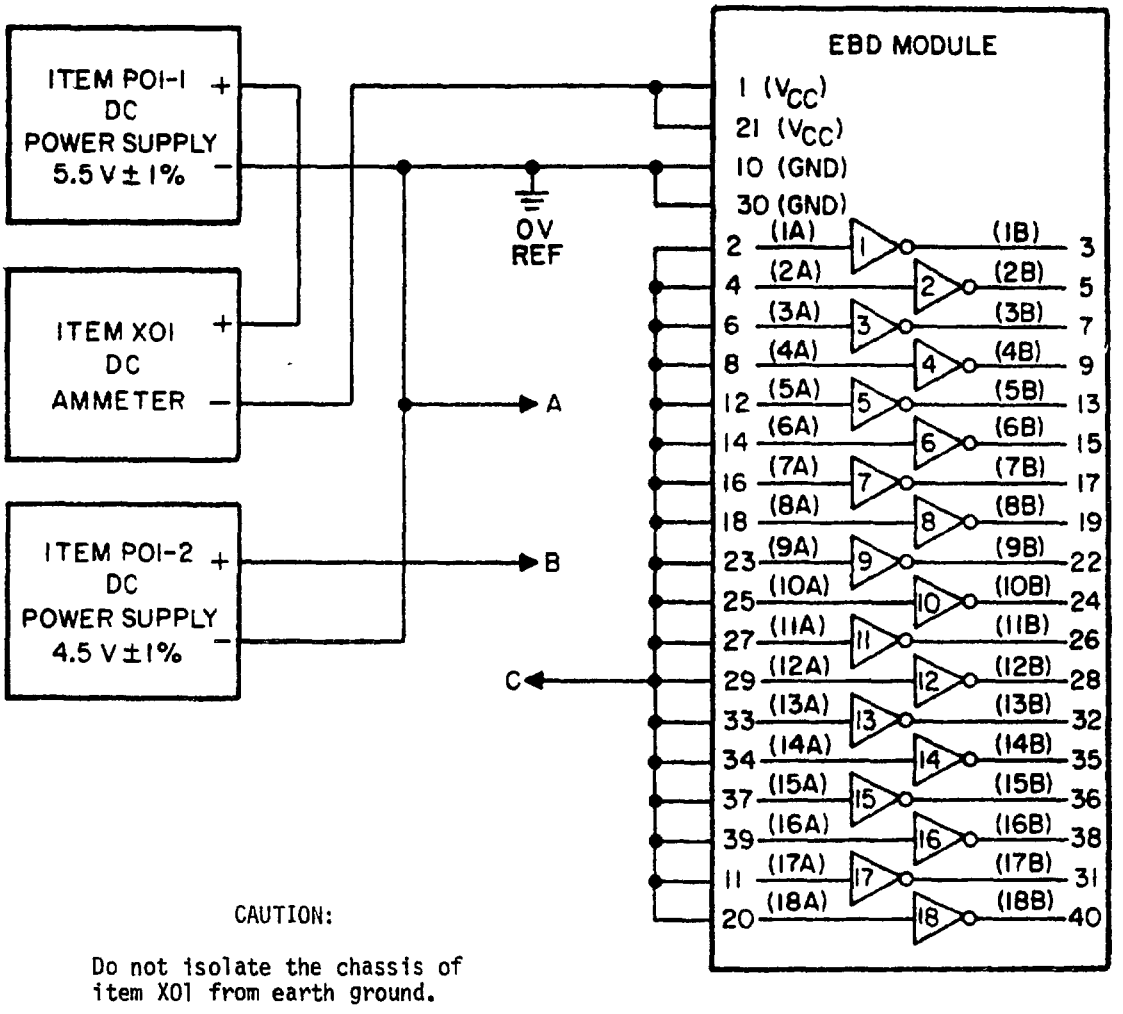


FIGURE 14. Power supply current (low-level and high-level outputs) test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1B Transition time (t_{TLH}), C_L = 50 pF	2	3	4	45	ns	15,16
M2	2B "	4	5	"	"	"	"
M3	3B "	6	7	"	"	"	"
M4	4B "	8	9	"	"	"	"
M5	5B "	12	13	"	"	"	"
M6	6B "	14	15	"	"	"	"
M7	7B "	16	17	"	"	"	"
M8	8B "	18	19	"	"	"	"
M9	9B "	23	22	"	"	"	"
M10	10B "	25	24	"	"	"	"
M11	11B "	27	26	"	"	"	"
M12	12B "	29	28	"	"	"	"
M13	13B "	33	32	"	"	"	"
M14	14B "	34	35	"	"	"	"
M15	15B "	37	36	"	"	"	"
M16	16B "	39	38	"	"	"	"
M17	17B "	11	31	"	"	"	"
M18	18B "	20	40	"	"	"	"
M19	1B Transition time (t_{TLH}), C_L = 150 pF	2	3	10	90	"	"
M20	2B "	4	5	"	"	"	"
M21	3B "	6	7	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M23	5B Transition time (t_{TLH}), C_L = 150 pF	12	13	10	90	ns	15,16
M24	6B "	14	15	"	"	"	"
M25	7B "	16	17	"	"	"	"
M26	8B "	18	19	"	"	"	"
M27	9B "	23	22	"	"	"	"
M28	10B "	25	24	"	"	"	"
M29	11B "	27	26	"	"	"	"
M30	12B "	29	28	"	"	"	"
M31	13B "	33	32	"	"	"	"
M32	14B "	34	35	"	"	"	"
M33	15B "	37	36	"	"	"	"
M34	16B "	39	38	"	"	"	"
M35	17B "	11	31	"	"	"	"
M36	18B "	20	40	"	"	"	"
N1	1B Transition time (t_{THL}), C_L = 50 pF	2	3	1	10	"	"
N2	2B "	4	5	"	"	"	"
N3	3B "	6	7	"	"	"	"
N4	4B "	8	9	"	"	"	"
N5	5B "	12	13	"	"	"	"
N6	6B "	14	15	"	"	"	"
N7	7B "	16	17	"	"	"	"

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TABLE IV. Transition time (t_{TLH} and t_{THL}) tests - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N9	9B Transition time (t_{THL}), C_L = 50 pF	23	22	1	10	ns	15,16
N10	10B "	25	24	"	"	"	"
N11	11B "	27	26	"	"	"	"
N12	12B "	29	28	"	"	"	"
N13	13B "	33	32	"	"	"	"
N14	14B "	34	35	"	"	"	"
N15	15B "	37	36	"	"	"	"
N16	16B "	39	38	"	"	"	"
N17	17B "	11	31	"	"	"	"
N18	18B "	20	40	"	"	"	"
N19	1B Transition time (t_{THL}), C_L = 150 pF	2	3	1	15	"	"
N20	2B "	4	5	"	"	"	"
N21	3B "	6	7	"	"	"	"
N22	4B "	8	9	"	"	"	"
N23	5B "	12	13	"	"	"	"
N24	6B "	14	15	"	"	"	"
N25	7B "	16	17	"	"	"	"
N26	8B "	18	19	"	"	"	"
N27	9B "	23	22	"	"	"	"
N28	10B "	25	24	"	"	"	"
N29	11B "	27	26	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N30	12B	Transition time (t_{THL}), C_L = 150 pF	29	28	1	15	ns	15,16
N31	13B	"	33	32	"	"	"	"
N32	14B	"	34	35	"	"	"	"
N33	15B	"	37	36	"	"	"	"
N34	16B	"	39	38	"	"	"	"
N35	17B	"	11	31	"	"	"	"
N36	18B	"	20	40	"	"	"	"

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TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P1	1B	Propagation delay time (t_{PLH}), C_L = 50 pF	2	3	4	40	ns	15,16
P2	2B	"	4	5	"	"	"	"
P3	3B	"	6	7	"	"	"	"
P4	4B	"	8	9	"	"	"	"
P5	5B	"	12	13	"	"	"	"
P6	6B	"	14	15	"	"	"	"
P7	7B	"	16	17	"	"	"	"
P8	8B	"	18	19	"	"	"	"
P9	9B	"	23	22	"	"	"	"
P10	10B	"	25	24	"	"	"	"
P11	11B	"	27	26	"	"	"	"
P12	12B	"	29	28	"	"	"	"
P13	13B	"	33	32	"	"	"	"
P14	14B	"	34	35	"	"	"	"
P15	15B	"	37	36	"	"	"	"
P16	16B	"	39	38	"	"	"	"
P17	17B	"	11	31	"	"	"	"
P18	18B	"	20	40	"	"	"	"
P19	1B	Propagation delay time (t_{PLH}), C_L = 150 pF	2	3	8	65	"	"
P20	2B	"	4	5	"	"	"	"
P21	3B	"	6	7	"	"	"	"
P22	4B	"	8	9	"	"	"	"

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P23	5B	Propagation delay time (t_{PLH}), C_L = 150 pF	12	13	8	65	ns	15,16
P24	6B	"	14	15	"	"	"	"
P25	7B	"	16	17	"	"	"	"
P26	8B	"	18	19	"	"	"	"
P27	9B	"	23	22	"	"	"	"
P28	10B	"	25	24	"	"	"	"
P29	11B	"	27	26	"	"	"	"
P30	12B	"	29	28	"	"	"	"
P31	13B	"	33	32	"	"	"	"
P32	14B	"	34	35	"	"	"	"
P33	15B	"	37	36	"	"	"	"
P34	16B	"	39	38	"	"	"	"
P35	17B	"	11	31	"	"	"	"
P36	18B	"	20	40	"	"	"	"
Q1	1B	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3	2	35	"	"
Q2	2B	"	4	5	"	"	"	"
Q3	3B	"	6	7	"	"	"	"
Q4	4B	"	8	9	"	"	"	"
Q5	5B	"	12	13	"	"	"	"
Q6	6B	"	14	15	"	"	"	"
Q7	7B	"	16	17	"	"	"	"

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

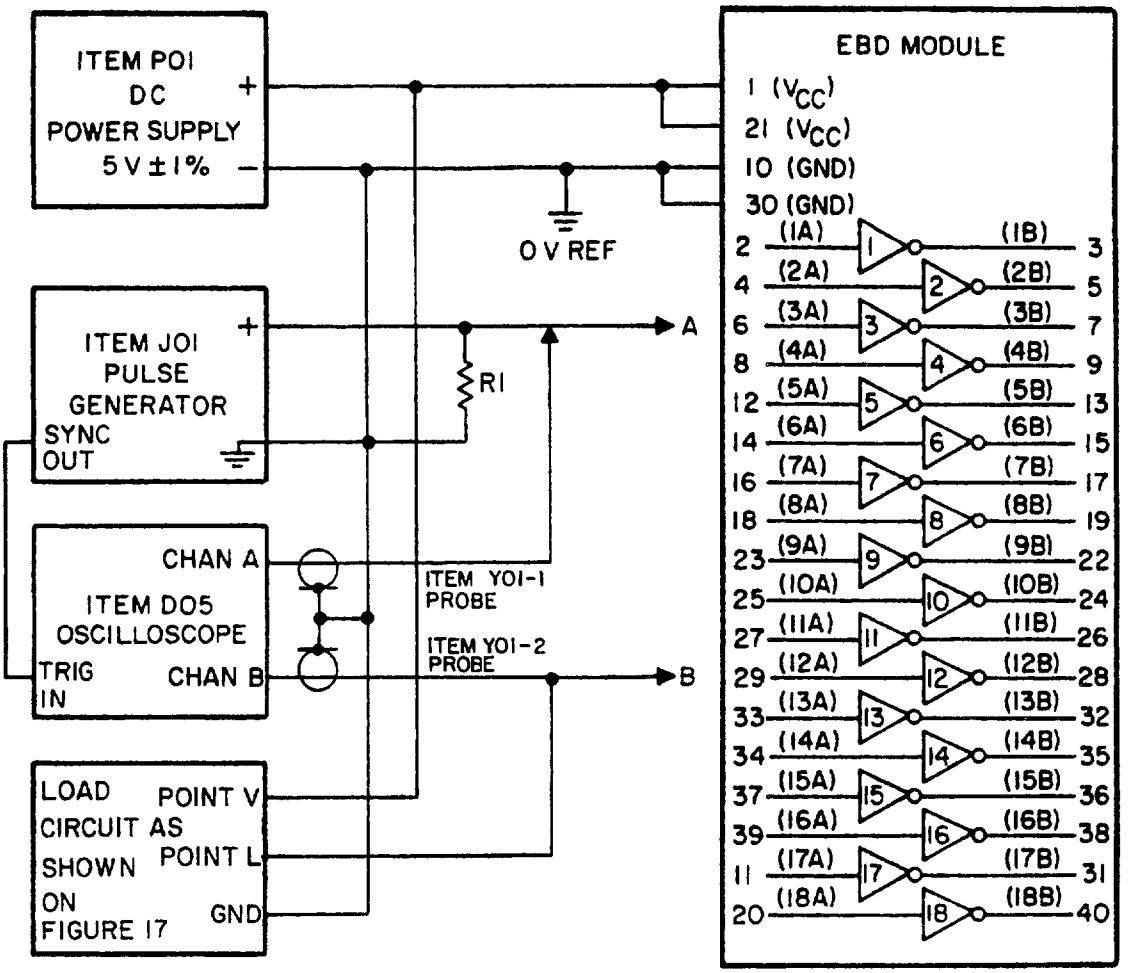
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q8	8B Propagation delay time (t_{PHL}), C_L = 50 pF	18	19	2	35	ns	15,16
Q9	9B "	23	22	"	"	"	"
Q10	10B "	25	24	"	"	"	"
Q11	11B "	27	26	"	"	"	"
Q12	12B "	29	28	"	"	"	"
Q13	13B "	33	32	"	"	"	"
Q14	14B "	34	35	"	"	"	"
Q15	15B "	37	36	"	"	"	"
Q16	16B "	39	38	"	"	"	"
Q17	17B "	11	31	"	"	"	"
Q18	18B "	20	40	"	"	"	"
Q19	1B Propagation delay time (t_{PHL}), C_L = 150 pF	2	3	4	50	"	"
Q20	2B "	4	5	"	"	"	"
Q21	3B "	6	7	"	"	"	"
Q22	4B "	8	9	"	"	"	"
Q23	5B "	12	13	"	"	"	"
Q24	6B "	14	15	"	"	"	"
Q25	7B "	16	17	"	"	"	"
Q26	8B "	18	19	"	"	"	"
Q27	9B "	23	22	"	"	"	"
Q28	10B "	25	24	"	"	"	"

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*TABLE XV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Q29	11B	Propagation delay time (t_{PHL}), C_L = 150 pF	27	26	4	50	ns	15,16
Q30	12B	"	29	28	"	"	"	"
Q31	13B	"	33	32	"	"	"	"
Q32	14B	"	34	35	"	"	"	"
Q33	15B	"	37	36	"	"	"	"
Q34	16B	"	39	38	"	"	"	"
Q35	17B	"	11	31	"	"	"	"
Q36	18B	"	20	40	"	"	"	"

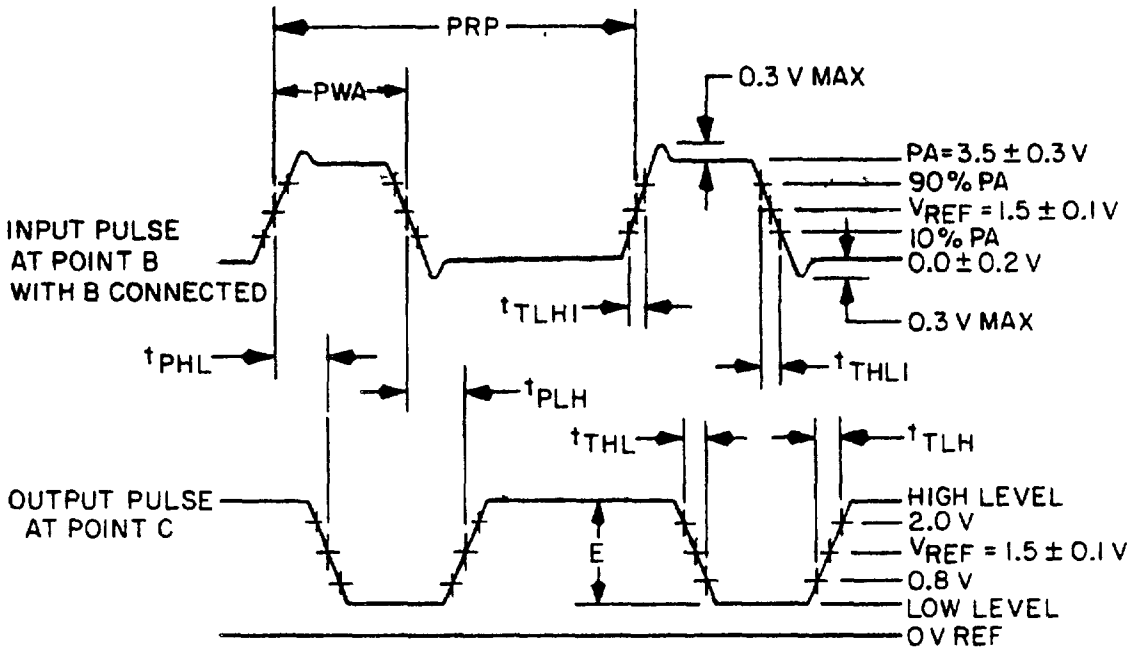
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NOTE: R1 = 51Ω ±5%, 1/4 W.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIV and XV.

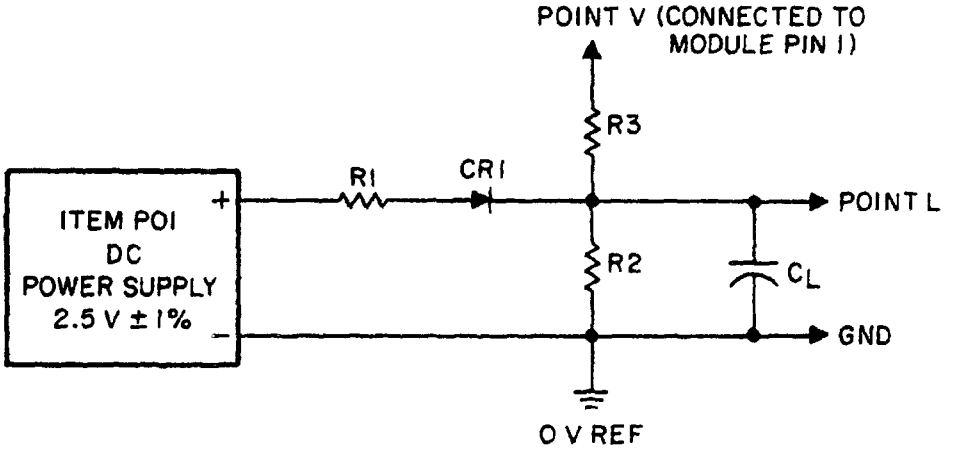
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to 0 V REF unless otherwise specified.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIV and XV.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow at point L of 16 mA ±5% with point L held at 0.4 V and point V held at 5 V.
3. R2 = 9.09 kΩ ±1%, 1/8 W.
4. R3 = 2.87 kΩ ±1%, 1/8 W.
5. Load A:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Load B:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
6. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

FIGURE 17. EBD logic load circuit.

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TABLE XVI. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1B Output crosstalk, low level	4,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	2	Open	3	0	1,250	mV	18,19
R2	2B "	2,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	4	"	5	"	"	"	"
R3	3B "	2,4,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	6	"	7	"	"	"	"
R4	4B "	2,4,6,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	8	"	9	"	"	"	"
R5	5B "	2,4,6,8,11,14,16,18,20, 23,25,27,29,33,34,37,39	12	"	13	"	"	"	"
R6	6B "	2,4,6,8,11,12,16,18,20, 23,25,27,29,33,34,37,39	14	"	15	"	"	"	"
R7	7B "	2,4,6,8,11,12,14,18,20, 23,25,27,29,33,34,37,39	16	"	17	"	"	"	"
R8	8B "	2,4,6,8,11,12,14,16,20, 23,25,27,29,33,34,37,39	18	"	19	"	"	"	"
R9	9B "	2,4,6,8,11,12,14,16,18,20, 25,27,29,33,34,37,39	23	"	22	"	"	"	"
R10	10B "	2,4,6,8,11,12,14,16,18, 20,23,27,29,33,34,37,39	25	"	24	"	"	"	"
R11	11B "	2,4,6,8,11,12,14,16,18, 20,23,25,29,33,34,37,39	27	"	26	"	"	"	"
R12	12B "	2,4,6,8,11,12,14,16,18, 20,23,25,27,33,34,37,39	29	"	28	"	"	"	"
R13	13B "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,34,37,39	33	"	32	"	"	"	"
R14	14B "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,37,39	34	"	35	"	"	"	"
R15	15B "	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,39	37	"	36	"	"	"	"

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TABLE XVI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R16	16B	Output crosstalk, low level	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,37	39	Open	38	0	1,250	mV	18,19
R17	17B	"	2,4,6,8,12,14,16,18,20, 23,25,27,29,33,34,37,39	11	"	31	"	"	"	"
R18	18B	"	2,4,6,8,11,12,14,16,18, 23,25,27,29,33,34,37,39	20	"	40	"	"	"	"
R19	1B	"	11,16,18,20,23,25,27,29, 33,34,37,39	2,4,6,8, 12,14	"	3	"	500	"	"
R20	2B	"	"	"	"	5	"	"	"	"
R21	3B	"	"	"	"	7	"	"	"	"
R22	4B	"	"	"	"	9	"	"	"	"
R23	5B	"	"	"	"	13	"	"	"	"
R24	6B	"	"	"	"	15	"	"	"	"
R25	7B	"	2,4,6,8,11,12,14,20,33, 34,37,39	16,18,23, 25,27,29	"	17	"	"	"	"
R26	8B	"	"	"	"	19	"	"	"	"
R27	9B	"	"	"	"	22	"	"	"	"
R28	10B	"	"	"	"	24	"	"	"	"
R29	11B	"	"	"	"	26	"	"	"	"
R30	12B	"	"	"	"	28	"	"	"	"
R31	13B	"	2,4,6,8,12,14,16,18,23, 25,27,29	11,20,33, 34,37,39	"	32	"	"	"	"
R32	14B	"	"	"	"	35	"	"	"	"
R33	15B	"	"	"	"	36	"	"	"	"
R34	16B	"	"	"	"	38	"	"	"	"
R35	17B	"	"	"	"	31	"	"	"	"
R36	18B	"	"	"	"	40	"	"	"	"

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TABLE XVI. Output crosstalk (low-level and high-level) test - Continued.

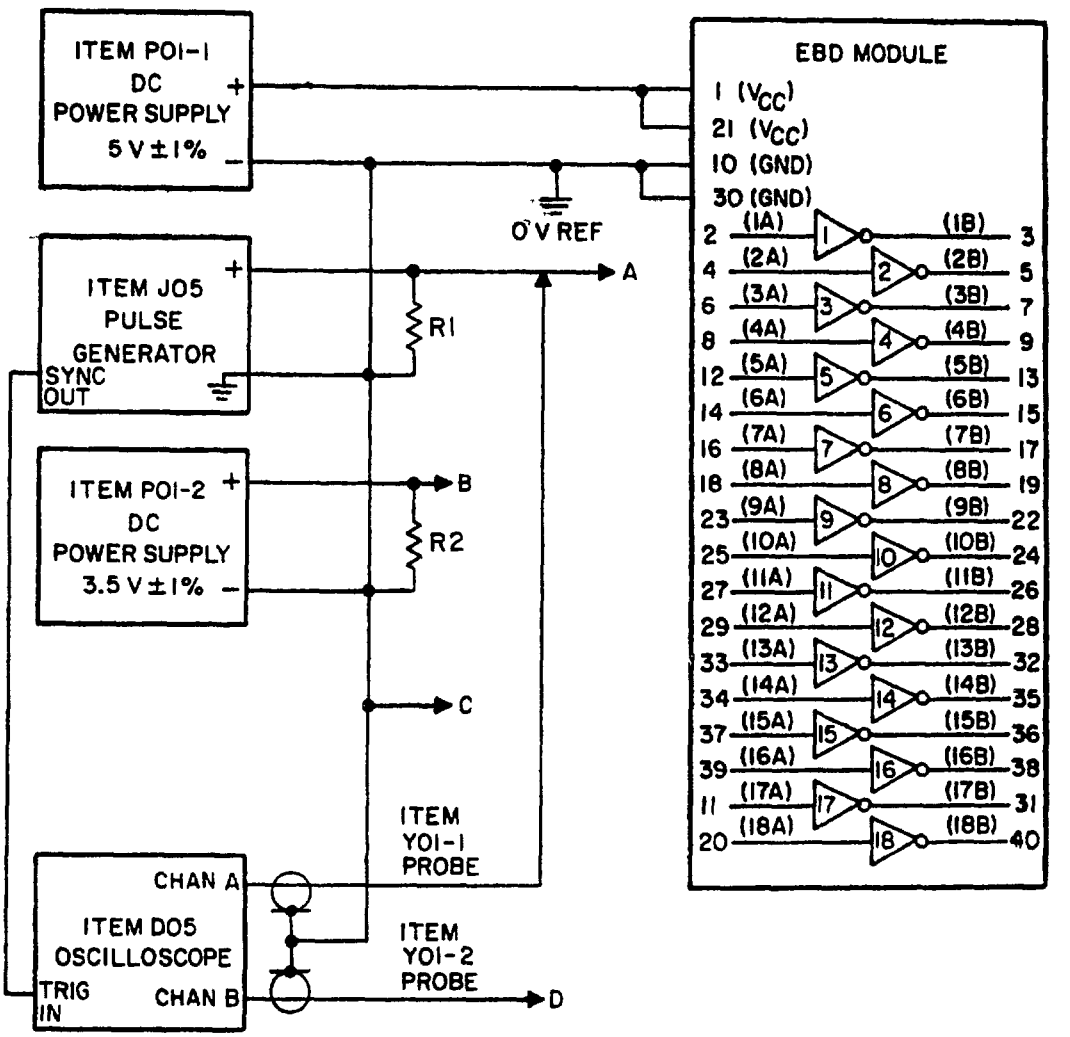
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1B	Output crosstalk, high level	4,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	Open	2	3	0	400	mV	18,19
S2	2B	"	2,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	"	4	5	"	"	"	"
S3	3B	"	2,4,8,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	"	6	7	"	"	"	"
S4	4B	"	2,4,6,11,12,14,16,18,20, 23,25,27,29,33,34,37,39	"	8	9	"	"	"	"
S5	5B	"	2,4,6,8,11,14,16,18,20, 23,25,27,29,33,34,37,39	"	12	13	"	"	"	"
S6	6B	"	2,4,6,8,11,12,16,18,20, 23,25,27,29,33,34,37,39	"	14	15	"	"	"	"
S7	7B	"	2,4,6,8,11,12,14,18,20, 23,25,27,29,33,34,37,39	"	16	17	"	"	"	"
S8	8B	"	2,4,6,8,11,12,14,16,20, 23,25,27,29,33,34,37,39	"	18	19	"	"	"	"
S9	9B	"	2,4,6,8,11,12,14,16,18, 20,25,27,29,33,34,37,39	"	23	22	"	"	"	"
S10	10B	"	2,4,6,8,11,12,14,16,18, 20,23,27,29,33,34,37,39	"	25	24	"	"	"	"
S11	11B	"	2,4,6,8,11,12,14,16,18, 20,23,25,29,33,34,37,39	"	27	26	"	"	"	"
S12	12B	"	2,4,6,8,11,12,14,16,18, 20,23,25,27,33,34,37,39	"	29	28	"	"	"	"
S13	13B	"	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,34,37,39	"	33	32	"	"	"	"
S14	14B	"	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,37,39	"	34	35	"	"	"	"
S15	15B	"	2,4,6,8,11,12,14,16,18, 20,23,25,27,29,33,34,39	"	37	36	"	"	"	"

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TABLE XVI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S16	16B	Output crosstalk, high level	2,4,6,8,11,12,14,16,18,20, 23,25,27,29,33,34,37	Open	39	38	0	400	mV	18,19
S17	17B	"	2,4,6,8,12,14,16,18,20, 23,25,27,29,33,34,37,39	"	11	31	"	"	"	"
S18	18B	"	2,4,6,8,11,12,14,16,18, 23,25,27,29,33,34,37,39	"	20	40	"	"	"	"

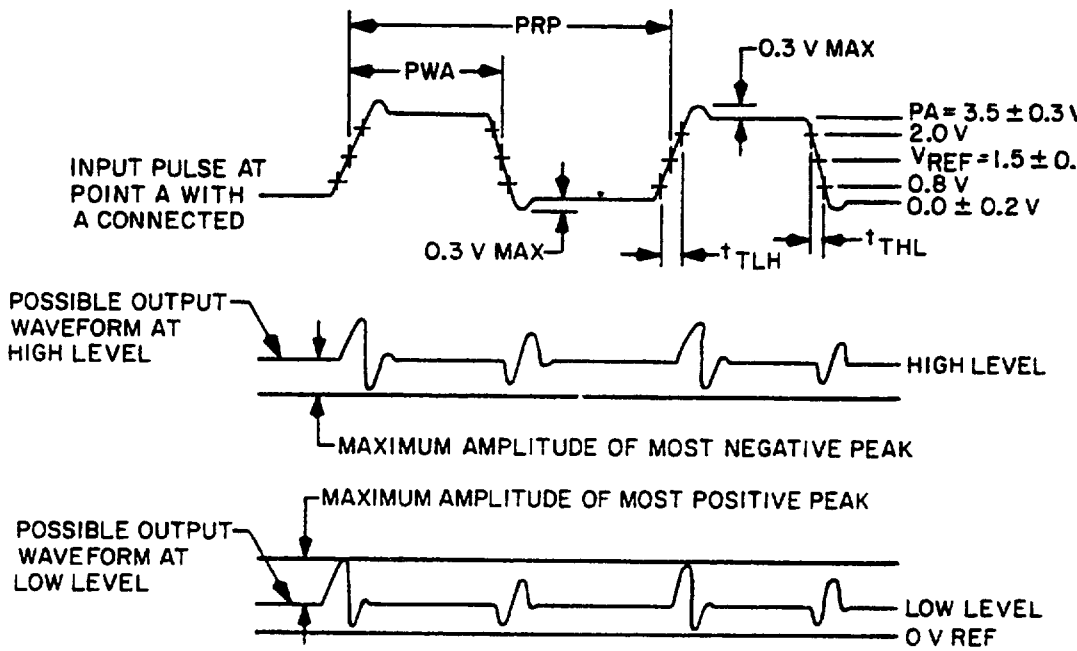
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/4 W.
2. Load each output simultaneously with load A as shown on figure 17 (C_L = 50 pF).

FIGURE 18. Output crosstalk (low-level and high-level) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   For each test condition with the module connected to item J05, adjust item J05 for the required input pulse parameters.
   Verify the input pulse parameters at each input pin before making each measurement.
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. All voltages are referenced to 0 V REF unless otherwise specified.

FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for table

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TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	20,21

TABLE XVIII. Power supply filter capacitance test.

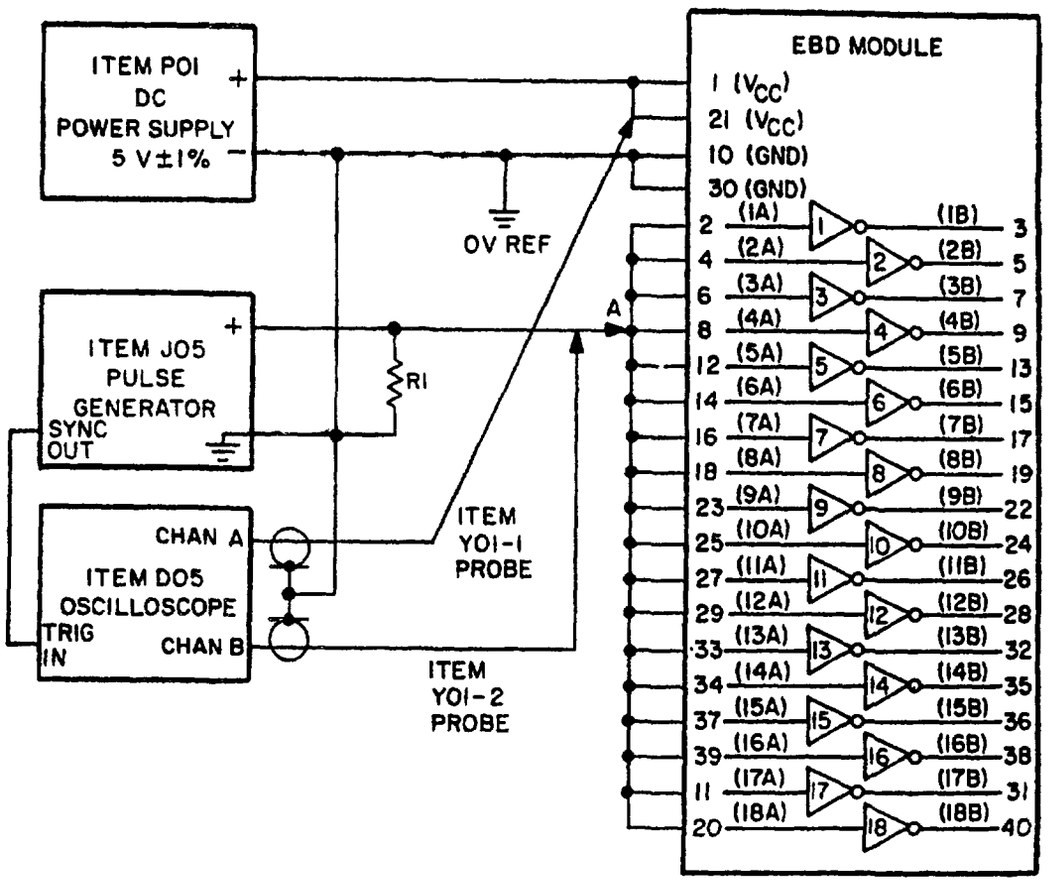
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1,21	1.6	3.0	µF	22

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TABLE X... Input capacitance tes..

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1A Input capacitance	2	Open	2	15	pF	22
V2	2A "	4	"	"	"	"	"
V3	3A "	6	"	"	"	"	"
V4	4A "	8	"	"	"	"	"
V5	5A "	12	"	"	"	"	"
V6	6A "	14	"	"	"	"	"
V7	7A "	16	"	"	"	"	"
V8	8A "	18	"	"	"	"	"
V9	9A "	23	"	"	"	"	"
V10	10A "	25	"	"	"	"	"
V11	11A "	27	"	"	"	"	"
V12	12A "	29	"	"	"	"	"
V13	13A "	33	"	"	"	"	"
V14	14A "	34	"	"	"	"	"
V15	15A "	37	"	"	"	"	"
V16	16A "	39	"	"	"	"	"
V17	17A "	11	"	"	"	"	"
V18	18A "	20	"	"	"	"	"

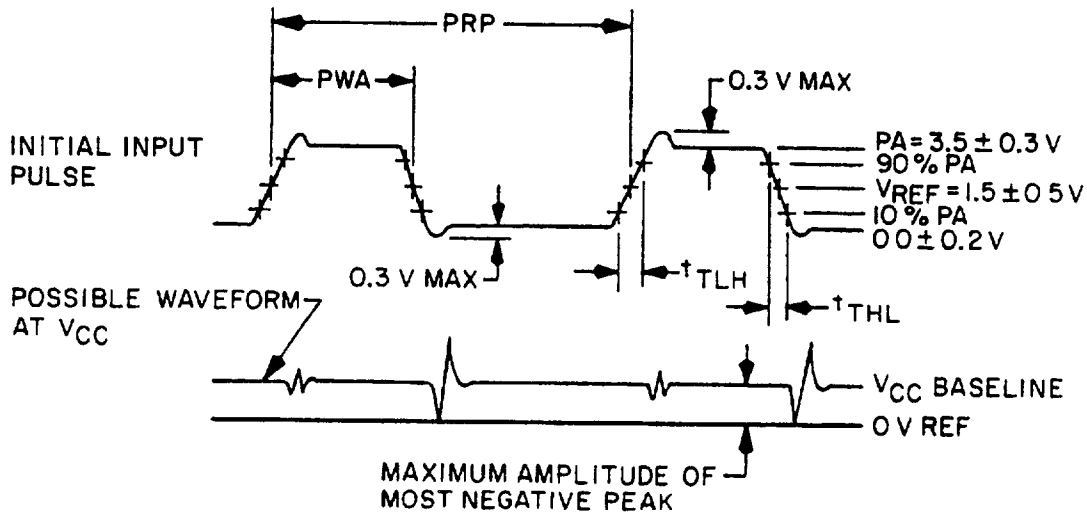
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NOTE: R1 = 51Ω ±5%, 1/4 W.

FIGURE 20. Power supply transient amplitude test circuit for table XVII.

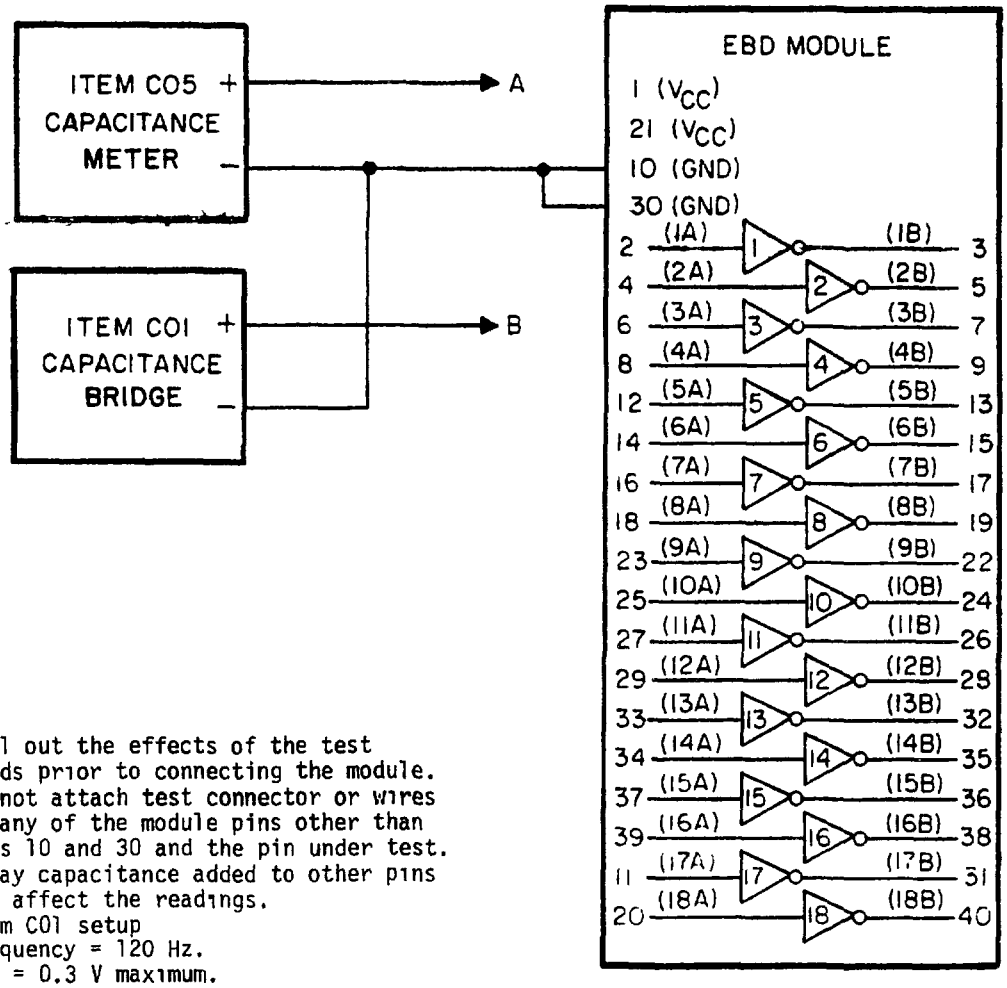
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NOTES:
1. Input pulse parameters:
   With the module connected to item J05, adjust item J05 for the required input pulse parameters.
   $t_{THL} = t_{TLH} = 15 \pm 2$ ns.
   PWA = 500 ±10 ns.
   PRP = 1,000 ±20 ns.
2. All voltages are referenced to 0 V REF, unless otherwise specified.

FIGURE 21. Power supply transient amplitude test circuit waveforms for table XVII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 22. Power supply filter and input capacitance test circuit for tables XVIII and XIX.

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